

[**N.B.** – The figures of the right margin indicate full marks. Read the stems carefully and answer the associated questions. Answer any **THREE** questions taking at least ONE question from each group]

Group - A

1. In a class of 40 students, 23 passed in Mathematics, 25 passed in English, and 10 failed in both the subjects. A student is randomly selected.
- (a) What does $P(A) = 0$ mean?

(b) Find the probability that the student passed in at least one subject.

(c) What is the probability that the student passed in both the subjects?

(d) Determine the probability that the student passed in only one subject.
2. The monthly electricity bills (in dollars) of 5 households are 45, 60, 55, 70, and 50. A researcher claims that if random samples of size 3 are drawn from these bills without replacement, the sample mean will provide an unbiased estimate of the population mean.

(a) Define a sample in this context.

(b) State two drawbacks of conducting a census.

(c) Compute the population variance from the given data.

(d) Verify the researcher’s claim regarding unbiasedness of the sample mean.
- Group - B
3. The joint probability function of two random variables X and Y is given by:
- $$P(X,Y) = \frac{x+y+1}{42}; \quad x = 0, 1, 2; \quad y = 0, 1, 2, 3$$
- (a) What is a discrete random variable?

(b) What are the required properties of a probability distribution?

(c) Calculate the marginal probability $P(Y)$.

(d) Determine $P(Y|X = 1)$ and $P(Y|X = 0)$.

4. The population of a country is growing exponentially at an annual rate of 2.5%.

(a) How does the arithmetic growth increase?

(b) What is the key difference between exponential growth and geometric growth?

(c) How many years will it take for the population to double under geometric growth and exponential growth?

(d) If the current population is 8 million, after 56 years, how many ****additional**** people will there be if the population grows exponentially instead of arithmetically?
- AAM–
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